How varied are stellar populations in the Solar vicinity?

Exploring GCNS and GALAH DR4 with t-SNE

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Abstract

We select a set of stars in the Solar vicinity observed by the Gaia Catalogue of Nearby Stars (GCNS; which comprises approximately 330k stars within 100 pc of the Sun) [2] that have also been followed-up with high-resolution spectroscopy by the GALactic Archaeology with HERMES (GALAH) DR4 survey [1]. In this context, the visualization of multidimensional data with potentially non-linear relationships remains a significant challenge. Our goal is to verify whether t-distributed Stochastic Neighbor Embedding (t-SNE) can retrieve the distinct stellar population properties in an unsupervised manner (effectively by clustering similar stars together), and to explore whether peculiar stars, such as emission-line objects, stand apart in the map (outlier detection itself is a later step). Hence, our study focuses on exploring how the t-SNE captures the main spectral features of the ∼5 000 stars in our sample. We tested and validated the t-SNE projection through spectral parameter-based color-mapping, which revealed a smooth gradient across the projected data, demonstrating that the algorithm successfully preserves the intrinsic physical correlations between several stellar parameters, such as $T_{\rm eff}$, $\log g$, and [Fe/H]. Furthermore, we examined the perplexity hyperparameter as a weighting factor between manifold scales, noting that a value of 30 increases sensitivity to local structures, whereas a value of 50 prioritizes the preservation of the global topology. Our preliminary results indicate the presence of a small detached sub-cluster, which maintains a different internal parameter gradient, distinguishing it from the main stellar group as a unique population within the local manifold, which requires further investigation.

Context: our previous work

A first, catalogue-level look at the stars common to the Gaia Catalogue of Nearby Stars (GCNS) and GALAH DR4 [8]. Now: does structure emerge from the spectra alone, with no catalogue properties as labels?

- **6 198** — stars in common to GCNS and GALAH DR4
- **FGK** — main-sequence dominated, a few A types; $T_{\rm eff}$ 3 000–8 000 K
- **1.5 Gyr** — median age (0.10–14.79 Gyr)
- **-0.19 dex** — median [Fe/H]: slightly metal-poor against the Sun
- **Thick disk** — with some probably halo members too α-enhanced

Methods: the pipeline, step by step

1 Sample — GCNS × GALAH DR4: 6 198 stars within 100 pc [1,2]

2 Spectra — native pixels, 4 CCDs (detectors) = 14 245 px; 894 lack a full set; 5 304 mapped

3 Projection — on the 5 304 stars with all 4 CCDs: cosine, no PCA [7]; t-SNE [3,6] and UMAP [4]

A 4 Colour, flag — **B** cuts: S/N and $T_{\rm eff}$; **C** colour: $T_{\rm eff}$, $\log g$, [Fe/H]; **D** split by flag

5 Emission — pick the emission bit; HDBSCAN [5] splits it into subclusters

E 6 Settings — same stars under t-SNE perplexity 50 and UMAP 15, 30

F 7 Spectra check — emitters vs quiet dwarfs, line by line, to test the flag

PCA vs t-SNE & UMAP. PCA (linear) blurs subtle differences between spectra; t-SNE and UMAP (non-linear) place similar spectra close together in 2D.

Perplexity. How many neighbours each star "looks at" when the map is built: 30 favours local structure, 50 the global topology (UMAP: 15 vs 30).

Overlap & distance correlation. Overlap: share of each star's 15 nearest neighbours kept in 2D (local). Distance correlation: Spearman of pairwise distances (global).

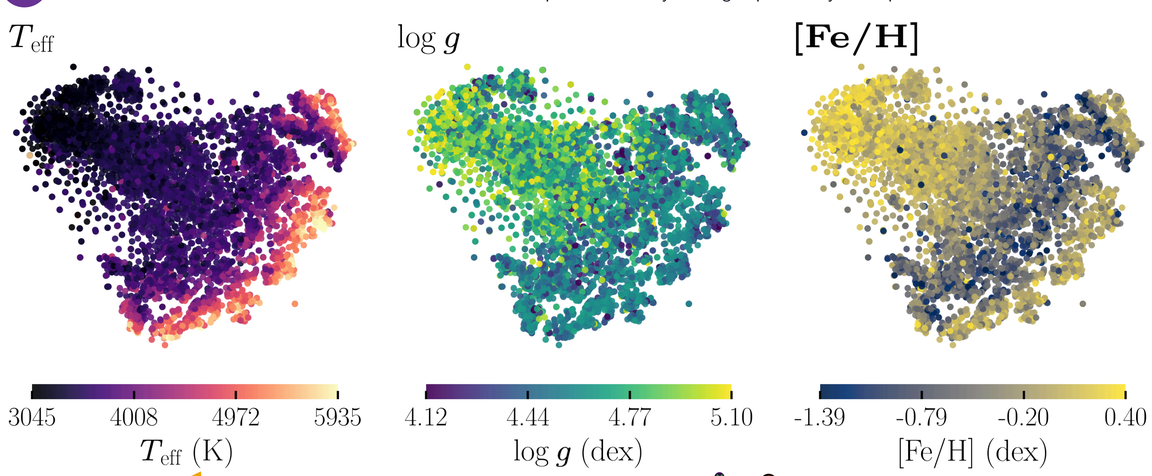
HDBSCAN (clustering). Finds dense groups of any shape without fixing their number; stars outside any dense group stay unassigned.

Aims. Explore the Solar neighbourhood with the full spectra, pixel by pixel: does structure emerge without labels, and how robust is it?

Results: what the maps and spectra show

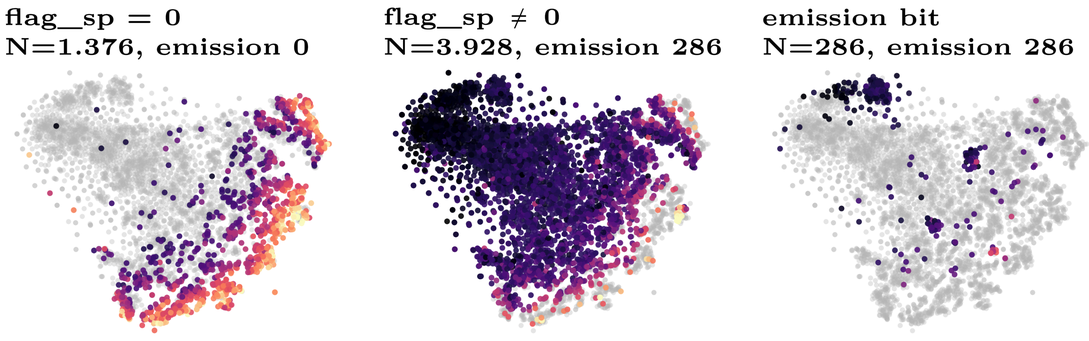
C Colouring the same map

Pixels only: $T_{\rm eff}$, $\log g$ and [Fe/H] never enter the algorithm; they only colour the map, to test whether spectral similarity alone groups stars by atmosphere.



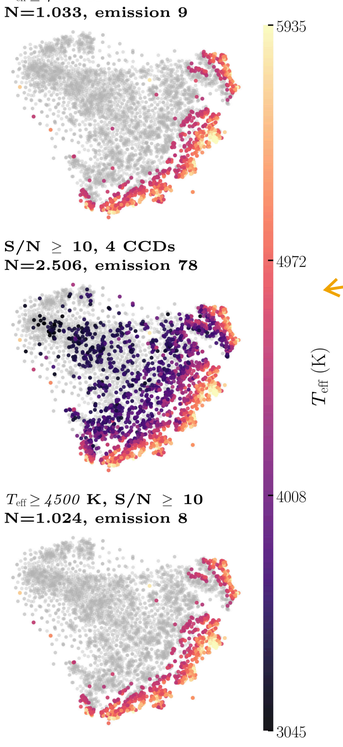
D Select by flag

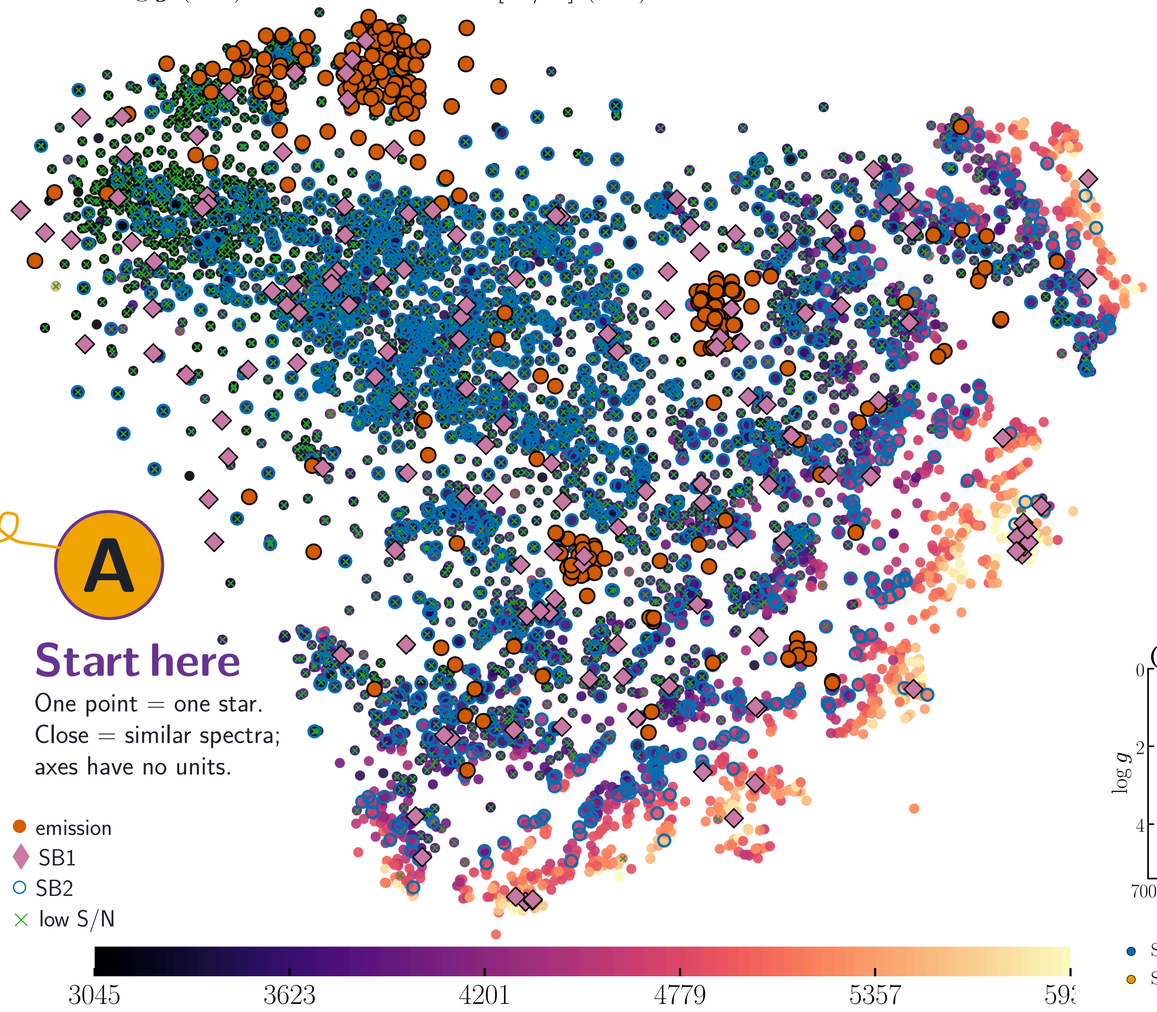
`flag_sp` is GALAH's spectrum-quality code (0 = no warning). Bit 1: emission; 2: a CCD detector is missing; 4 and 8: SB1 and SB2, binary-star candidates whose spectrum shows one (SB1) or two (SB2) sets of lines; 2048: low S/N (a noisy spectrum). Catalogue labels, not confirmed binaries.



B Cuts

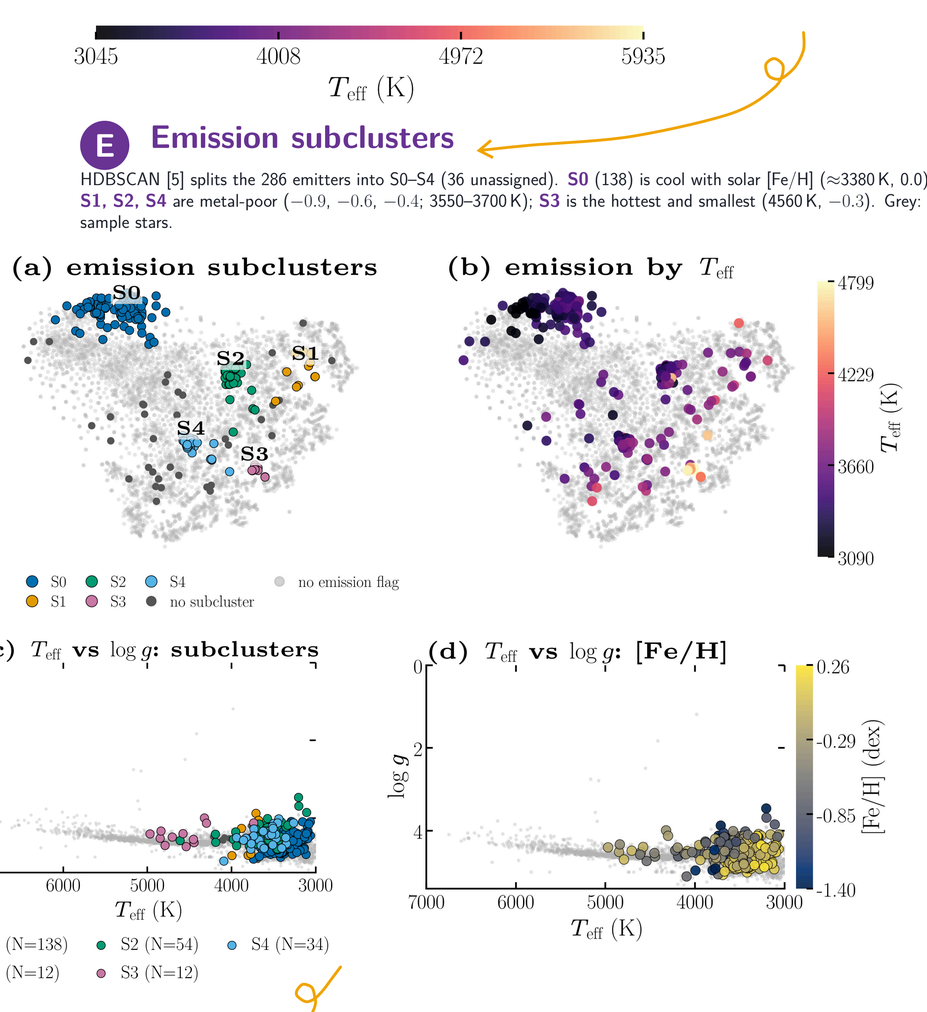
Cuts hide stars to test robustness: $T_{\rm eff} \geq 4500$ K (hotter only), S/N ≥ 10 in all 4 CCDs (cleanest spectra), and both.





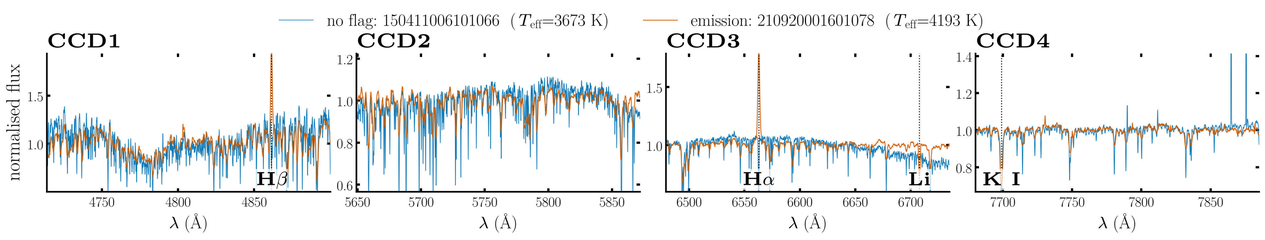
E Emission subclusters

HDBSCAN [5] splits the 286 emitters into S0–S4 (36 unassigned). **S0** (138) is cool with solar [Fe/H] (≈3380 K, 0.0); **S1, S2, S4** are metal-poor (-0.9, -0.6, -0.4; 3550–3700 K); **S3** is the hottest and smallest (4560 K, -0.3). Grey: all sample stars.



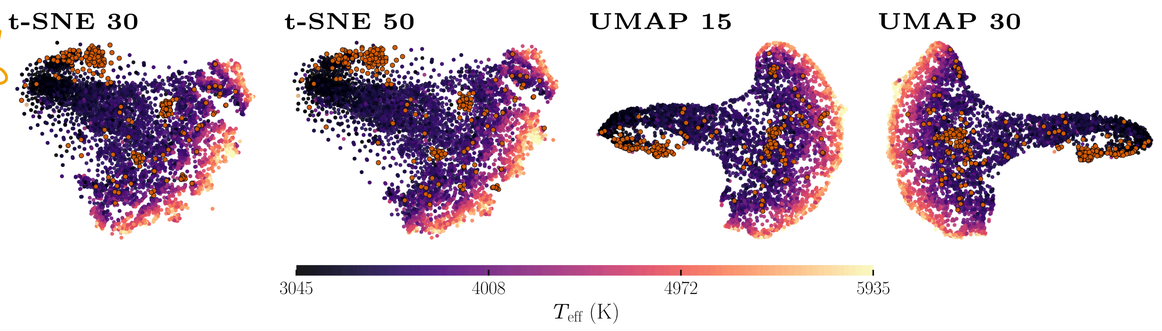
G Spectra: emitter vs quiet dwarf

Why look at spectra? A flag is only a catalogue label; the spectrum shows what the star is actually doing. A real emitter has Hα (and Hβ) filled in or in emission, which a quiet dwarf of similar $T_{\rm eff}$ lacks, so the line profile separates genuine chromospheric activity from a spurious flag.



F Other settings: does it hold?

Same stars under t-SNE perplexity 30 and 50 and UMAP with 15 and 30 neighbours (parameters defined in Context). **Result:** emitters (orange) stay grouped in all four; 30 to 50 raises distance correlation (0.38 to 0.43); UMAP reaches 0.44–0.49 with less overlap (0.25–0.21 vs 0.37).



Key findings

- $T_{\rm eff}$ recovered without labels in every setting.
- Cuts hide stars on one map (no re-projection); 52% carry the catalogue low-S/N flag, separate from our S/N ≥ 10 cut.
- Candidates, not classes: HDBSCAN settings and cuts not yet swept.
- Emitters stay grouped in all settings while local and global structure vary; the flagged example shows a real Hα peak.
- Next: all GALAH DR4 (905 415 spectra).

References

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